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M.Sc. QUANTITATIVE FINANCE
SEMESTER III
LECTURE NOTES ON
Machine Learning in Quantitative Finance

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OCTOBER 2026

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Chapter 1

Machine Learning Foundations & Data Preprocessing

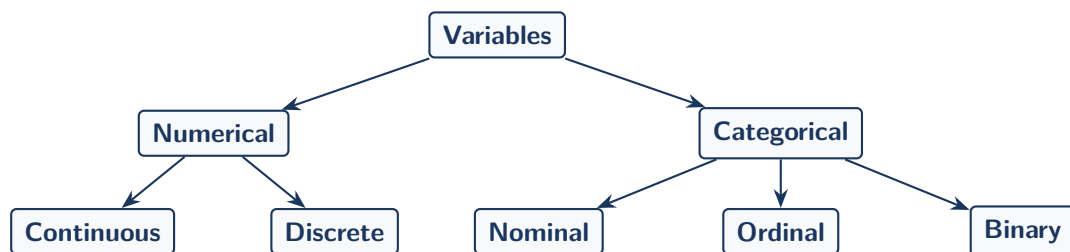
1.1 Challenges in Machine Learning

Machine learning applications in financial engineering face several structural and regulatory constraints:

- **Requires Large, Clean Data:** Models demand substantial volumes of high-quality, cleansed historical records to generalize effectively.
- **Risk of Overfitting:** Complex models risk fitting noise and idiosyncratic market movements rather than genuine underlying signals.
- **Interpretability (“Black Box”):** Advanced algorithms often lack transparent mechanistic explanations, creating auditing hurdles.
- **Regulatory & Ethical Concerns in Finance:** Strict regulatory compliance mandates model explainability, fairness, and accountability in financial decision-making.

1.2 Types of Variables

Variables encountered in machine learning datasets are broadly categorized into numerical and categorical types:

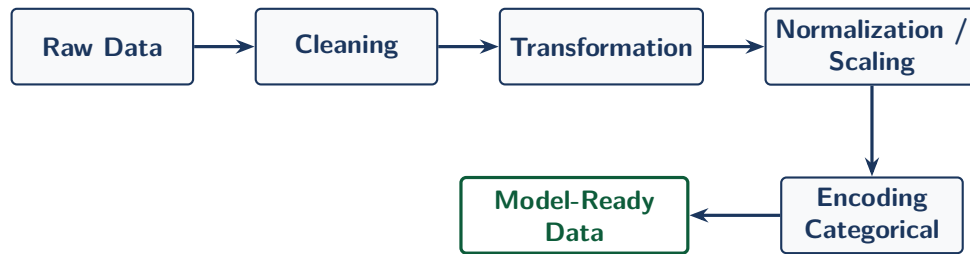


1.3 Data Preprocessing Pipeline

The standard end-to-end data preparation workflow transforms raw financial inputs into model-ready datasets:

Table 1.1: Taxonomy of Data Types and Financial Examples

Type	Subtype	Meaning	Finance Example
Numerical	Continuous	Any value in a continuous range	Stock price, portfolio return, volatility
Numerical	Discrete	Countable whole numbers	Number of trades per day, number of defaults
Categorical	Nominal	No natural ordering among categories	Sector: Banking, IT, Pharma
Categorical	Ordinal	Natural hierarchical order exists	Credit Rating: AAA > AA > A > BBB
Categorical	Binary	Exactly two distinct categories	Loan Default: Yes / No



- **Cleaning:** Removes duplicate records, corrects erroneous entries, and handles missing observations.
- **Transformation:** Converts raw figures into mathematically suitable modeling formats (e.g., calculating log returns from sequential stock prices).
- **Normalization / Scaling:** Brings numerical features across heterogeneous magnitudes into a common comparable scale.
- **Encoding:** Converts non-numeric categorical attributes into numerical indicator representations.

1.3.1 Scaling & Normalization

Features measured in disparate units must be rescaled to ensure distance-based and gradient-based algorithms treat features proportionately:

Min-Max Scaling

Rescales feature values into the bounded closed interval $[0, 1]$:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1.1)$$

Z-Score Standardization

Standardizes observations using empirical sample mean and sample standard deviation:

$$Z = \frac{x - \mu}{\sigma} \quad (1.2)$$

1.4 Train-Test-Validation Splitting

Datasets are partitioned into distinct subsets to evaluate model performance on unseen data, verifying whether the algorithm generalizes rather than memorizing training samples:

- **Training Set:** Used to fit model parameters and learn patterns.
- **Validation Set:** Used during model development to tune hyperparameters and guide model selection.
- **Testing Set:** Used strictly for final unbiased model performance evaluation.

1.4.1 Common Partitioning Ratios

- **Train-Test Splits:** 80% – 20% or 70% – 30%.
- **Train-Validation-Test Splits:** 70% – 15% – 15%.

1.4.2 K-Fold Cross-Validation

In k -fold cross-validation:

1. The training dataset is partitioned into k equal parts (“folds”).
2. The model is trained on $k - 1$ folds and validated on the remaining fold.
3. This procedure is repeated k times, rotating the held-out validation fold sequentially across all splits.

Benefits & Common Choices

- Provides a significantly more robust estimate of true out-of-sample model performance.
- Reduces vulnerability to an anomalous single lucky or unlucky split.
- Common configurations: $k = 5$ or $k = 10$.

1.4.3 Data Leakage Prevention

Data Leakage Warning

Never scale or normalize the entire dataset prior to splitting.

If sample mean μ and standard deviation σ are calculated over the aggregate dataset, information from the test set leaks directly into the training stage.

Mandatory Pipeline Sequence:

1. Split the raw dataset first.
2. Fit the scaler/imputer exclusively on training data.
3. Transform validation and test data using the saved training parameters.

1.4.4 Splitting Financial Time-Series Data

For time-series data like stock prices, a purely random split is fundamentally problematic because it mixes past and future observations. Sequential financial records must strictly use **chronological splits**.

1.5 Missing Data Mechanisms & Imputation

1.5.1 Missingness Mechanisms

Table 1.2: Mechanisms of Missing Data and Practical Financial Examples

Type	Meaning	Finance Example
MCAR	Missing Completely at Random: Missingness is entirely independent of observed and unobserved variables.	Trade entry terminal intermittently experiences random network packet drops logging trades.
MAR	Missing at Random: Missingness is statistically related to other observed features in the dataset.	Younger retail banking customers are systematically more likely to leave the optional income field blank.
MNAR	Missing Not at Random: Missingness is directly correlated with the actual missing value itself.	Highly indebted distressed borrowers deliberately omit outstanding secondary credit obligations.

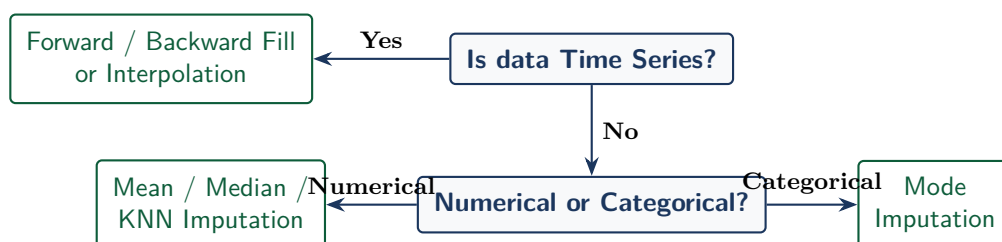
1.5.2 Imputation Methods

Common techniques for replacing missing data include: • Mean / Median / Mode, • K-Nearest Neighbors (KNN) Imputation, • Forward Fill, • Backward Fill, and • Interpolation.

Table 1.3: Advantages and Disadvantages of Imputation Methods

Method	Advantages	Disadvantages
Mean / Median	Simple, fast to compute	Distorts sample variance; ignores inter-feature relationships.
Mode	Directly applicable to categorical data	Can artificially over-represent the most common category.
KNN	Exploits multivariate relationships between features	Computationally expensive on large financial datasets.
Forward / Backward Fill	Well-suited for time-series sequences	Assumes most recent observed value remains representative.
Interpolation	Produces smooth estimates capturing trends	Unsuitable for non-sequential, unordered data.

1.5.3 Decision Framework for Selecting Imputation Methods



1.6 Imbalanced Categorical Data

Class imbalance arises when one category of the target variable appears far more frequently than other categories:

Financial Example

In credit card fraud detection:

- Genuine Transactions: 99.9%
- Fraudulent Transactions: 0.1%

A naive model that blindly predicts “no fraud” every single time achieves 99.9% raw accuracy, yet is entirely useless. Thus, **accuracy alone is fundamentally misleading**.

1.6.1 Consequences of Class Imbalance

- Biases model parameters heavily toward majority class patterns.
- Renders overall accuracy metric uninformative for decision-makers.

1.6.2 Remedial Solutions

- **Oversampling**: Duplicating or synthetically generating minority class instances.
- **Undersampling**: Down-sampling majority class observations.
- **SMOTE**: Synthetic Minority Over-sampling Technique, interpolating new instances along minority feature lines.

1.7 Model Evaluation Metrics

For binary classification problems, predictions and actual outcomes are summarized in a confusion matrix:

Table 1.4: Binary Classification Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Precision

Of all instances predicted as positive, how many were genuinely correct?

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1.3)$$

Recall

Of all actual positive instances in the population, how many did the model identify?

$$\text{Recall} = \frac{TP}{TP + FN} \quad (1.4)$$

F_1 -Score

Harmonic mean of Precision and Recall, balancing false alarms against missed detections:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.5)$$

1.8 Mistakes to be Avoided in ML Pipelines

1. **Global Scaling Before Splitting:** Scaling or normalizing the entire dataset prior to train/test partitioning causes data leakage.
2. **Blind Mean Imputation on MNAR Data:** Imputing missing values with unconditional means when data is Missing Not at Random injects substantial estimation bias.
3. **Ignoring Class Imbalance:** Relying exclusively on overall classification accuracy in skewed datasets conceals poor minority detection.
4. **Unchecked One-Hot Encoding on High-Cardinality Variables:** Creating dummy variables for high-cardinality categorical features leads to severe dimensionality inflation.
5. **Failing to Shuffle Non-Time-Series Data:** Neglecting random shuffling before partitioning standard cross-sectional data risks clustering-induced bias.

1.9 Key Takeaways

Core Principles of ML Data Preparation

- i) **Pre-processing dominates:** Data preprocessing accounts for 60% – 80% of total machine learning workflow effort.
- ii) **Always split first:** Partition data into training, validation, and test sets before performing any scaling or modeling.
- iii) **Diagnose missingness:** Determine whether missing values are MCAR, MAR, or MNAR before selecting an imputation strategy.
- iv) **Address class imbalance:** Use SMOTE, oversampling, or undersampling; accuracy alone is never sufficient.
- v) **Respect temporal order:** For financial and time-series data, strictly employ chronological splits and forward-looking imputation to prevent look-ahead bias.

Chapter 2

K-Nearest Neighbors (KNN)

2.1 Introduction to KNN

Definition 2.1.1: K-Nearest Neighbors

K-Nearest Neighbors (KNN) is a non-parametric, distance-based supervised machine learning algorithm used for both classification and regression tasks.

2.1.1 Core Algorithmic Characteristics

- **Supervised Learning:** Operates on labeled training datasets to predict continuous targets or discrete classes.
- **Dual Capability:** Utilized for classification (via majority voting) as well as regression (via local averaging).
- **Distance-Based Metric Method:** Assigns predictions purely based on geometric proximity in the feature space.
- **Lazy Learning / Instance-Based:** Does not construct an explicit internal model or parameter set during training; retains the entire training dataset and defers computation until inference time.

2.1.2 Algorithmic Workflow

The operational execution pipeline of KNN proceeds through four sequential stages:



2.2 Distance Metrics

Geometric closeness between query observations and stored instances is computed using formal metric distance functions.

2.2.1 Euclidean Distance

For two observations in a 2-dimensional feature space, $\mathbf{X} = (x_1, x_2)$ and $\mathbf{Y} = (y_1, y_2)$:

$$d(\mathbf{X}, \mathbf{Y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \quad (2.1)$$

For an arbitrary p -dimensional feature vector (or n features):

$$d(\mathbf{X}, \mathbf{Y}) = \sqrt{\sum_{j=1}^p (x_j - y_j)^2} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (2.2)$$

2.2.2 Manhattan Distance

The Manhattan distance (L_1 norm) measures rectilinear city-block distance across coordinates:

$$d(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^n |x_i - y_i| \quad (2.3)$$

- **For 2 Dimensions:** $d = |x_1 - x_2| + |y_1 - y_2|$
- **For 3 Dimensions:** $d = |x_1 - x_2| + |y_1 - y_2| + |z_1 - z_2|$

2.2.3 Minkowski Distance

The generalized Minkowski metric of order $p \geq 1$ encompasses both Manhattan and Euclidean distances:

$$d(\mathbf{X}, \mathbf{Y}) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p} \quad (2.4)$$

Table 2.1: Minkowski Distance Orders and Resulting Metrics

Order p	Distance Metric	Mathematical Formula
$p = 1$	Manhattan distance	$\sum x_i - y_i $
$p = 2$	Euclidean distance	$\sqrt{\sum (x_i - y_i)^2}$
$p \geq 1$	Minkowski distance	$\left(\sum x_i - y_i ^p \right)^{1/p}$

Where:

- n = number of dimensions (features)
- p = positive real number (the “order” of the distance metric)
- x_i, y_i = the i -th feature coordinates of the two compared observations

2.3 Worked Numerical Examples

2.3.1 Fruit Classification via Euclidean Distance

Example 2.3.1: Fruit Classification (Apple vs. Lemon)

A dataset contains observations evaluated on two physical attributes: *Sweetness* and *Crunchiness*.

Fruit Training Dataset

Point	Sweetness	Crunchiness	Class
<i>A</i>	7	7	Apple
<i>B</i>	8	6	Apple
<i>C</i>	3	8	Lemon
<i>D</i>	2	7	Lemon
<i>E</i>	6	8	Apple
<i>F</i>	1	9	Lemon

Task: Classify a new query observation $\mathbf{X} = (5, 7)$ using $k = 3$ and Euclidean distance.

Solution:

Step 1: Compute Euclidean distances from query point $\mathbf{X}(5, 7)$:

$$\text{To } A(7, 7) : d = \sqrt{(5 - 7)^2 + (7 - 7)^2} = \sqrt{4 + 0} = 2.00$$

$$\text{To } B(8, 6) : d = \sqrt{(5 - 8)^2 + (7 - 6)^2} = \sqrt{9 + 1} = \sqrt{10} \approx 3.16$$

$$\text{To } C(3, 8) : d = \sqrt{(5 - 3)^2 + (7 - 8)^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.23$$

$$\text{To } D(2, 7) : d = \sqrt{(5 - 2)^2 + (7 - 7)^2} = \sqrt{9 + 0} = 3.00$$

$$\text{To } E(6, 8) : d = \sqrt{(5 - 6)^2 + (7 - 8)^2} = \sqrt{1 + 1} = \sqrt{2} \approx 1.41$$

$$\text{To } F(1, 9) : d = \sqrt{(5 - 1)^2 + (7 - 9)^2} = \sqrt{16 + 4} = \sqrt{20} \approx 4.04$$

Step 2: Sort distances in ascending order:

Point	Distance	Class
<i>E</i>	1.41	Apple
<i>A</i>	2.00	Apple
<i>C</i>	2.23	Lemon
<i>D</i>	3.00	Lemon
<i>B</i>	3.16	Apple
<i>F</i>	4.04	Lemon

Step 3: Evaluate majority class for $k = 3$:

$$\{k = 3\} = \{E, A, C\}$$

- Apple: 2 votes (E, A)
- Lemon: 1 vote (C)

$\therefore \mathbf{X}$ is to be classified as **Apple**.

2.3.2 Fintech Stock Recommendation with Standardized Indicators

Example 2.3.2: Fintech Stock Recommendation (Buy, Hold, Avoid)

A fintech company builds a KNN system to recommend whether a stock is a **Buy**, **Hold**, or **Avoid** based on three financial indicators: Price-to-Earnings (P/E), Return on Equity (ROE), and Debt-to-Equity (D/E).

The training samples are provided after standardizing each indicator via $Z = \frac{x-\mu}{\sigma}$:

Standardized Financial Indicator Training Dataset

Stock	$P/E (Z)$	$ROE (Z)$	$D/E (Z)$	Class
S_1	-1.351	1.542	-1.184	Buy
S_2	-0.787	0.806	-0.858	Buy
S_3	-0.129	-0.299	-0.204	Hold
S_4	0.341	-0.668	0.286	Hold
S_5	1.750	-1.404	1.757	Avoid
S_6	1.093	-1.036	1.267	Avoid
S_7	-1.069	1.174	-1.021	Buy
S_8	0.153	-0.115	-0.041	Hold

Task: Predict the recommendation class for a new stock with standardized features:

$$\mathbf{X} = (-0.559, 0.437, -0.531)$$

using $k = 3$ and Euclidean distance.

Solution:

Step 1: Compute Euclidean distance for each stock:

$$d(\mathbf{X}, S_i) = \sqrt{(z_{1,X} - z_{1,S_i})^2 + (z_{2,X} - z_{2,S_i})^2 + (z_{3,X} - z_{3,S_i})^2}$$

Stock	$P/E (Z)$	$ROE (Z)$	$D/E (Z)$	Distance	Class
S_1	-1.351	1.542	-1.184	1.487	Buy
S_2	-0.787	0.806	-0.858	0.527	Buy
S_3	-0.129	-0.299	-0.204	0.939	Hold
S_4	0.341	-0.668	0.286	1.266	Hold
S_5	1.750	-1.404	1.757	3.760	Avoid
S_6	1.093	-1.036	1.267	2.874	Avoid
S_7	-1.069	1.174	-1.021	1.002	Buy
S_8	0.153	-0.115	-0.041	1.053	Hold

Step 2: Sorted Euclidean distance table:

Stock	Distance	Class
S_2	0.527	Buy
S_3	0.939	Hold
S_7	1.002	Buy
S_8	1.053	Hold
S_4	1.266	Hold
S_1	1.487	Buy
S_6	2.874	Avoid
S_5	3.760	Avoid

Step 3: Majority voting for $k = 3$:

$$\{k = 3\} = \{S_2, S_3, S_7\}$$

- Buy: 2 votes (S_2, S_7)
- Hold: 1 vote (S_3)
- Avoid: 0 votes

$\therefore$ The new point is to be classified as **Buy** for $k = 3$.

2.3.3 Bank Loan Approval via Manhattan Distance

Example 2.3.3: Bank Loan Approval Decision

A bank utilizes KNN to decide whether to **Approve** or **Reject** a loan application based on two features:

- Credit score (scaled by dividing by 100, such that $750 \implies 7.5$)
- Annual Income (in Rs. Lakhs)

Loan Applicant Training Dataset

Applicant	Credit Score	Annual Income (Rs. Lakhs)	Decision
A_1	7.5	8.0	Approve
A_2	6.8	6.0	Approve
A_3	6.0	4.0	Reject
A_4	5.5	3.0	Reject
A_5	7.2	7.0	Approve
A_6	5.8	3.5	Reject

Task: Predict the decision for a new applicant with Credit Score = 6.5 and Annual Income = 5.5 Lakhs, i.e., $\mathbf{X} = (6.5, 5.5)$, using $k = 3$ and Manhattan distance.

Solution:

Step 1: Compute Manhattan distances from query point $\mathbf{X}(6.5, 5.5)$:

$$\text{To } A_1(7.5, 8.0) : d = |6.5 - 7.5| + |5.5 - 8.0| = 1.0 + 2.5 = 3.5$$

$$\text{To } A_2(6.8, 6.0) : d = |6.5 - 6.8| + |5.5 - 6.0| = 0.3 + 0.5 = 0.8$$

$$\text{To } A_3(6.0, 4.0) : d = |6.5 - 6.0| + |5.5 - 4.0| = 0.5 + 1.5 = 2.0$$

$$\text{To } A_4(5.5, 3.0) : d = |6.5 - 5.5| + |5.5 - 3.0| = 1.0 + 2.5 = 3.5$$

$$\text{To } A_5(7.2, 7.0) : d = |6.5 - 7.2| + |5.5 - 7.0| = 0.7 + 1.5 = 2.2$$

$$\text{To } A_6(5.8, 3.5) : d = |6.5 - 5.8| + |5.5 - 3.5| = 0.7 + 2.0 = 2.7$$

Step 2: Sorted Manhattan distance ranking:

Applicant	Distance	Decision
A_2	0.8	Approve
A_3	2.0	Reject
A_5	2.2	Approve
A_6	2.7	Reject
A_4	3.5	Reject
A_1	3.5	Approve

Step 3: Majority vote for $k = 3$:

$$\{k = 3\} = \{A_2, A_3, A_5\}$$

- Approve: 2 votes (A_2, A_5)
- Reject: 1 vote (A_3)

$\therefore$ The new applicant $\mathbf{X} = (6.5, 5.5)$ is to be concluded as **Approve** when $k = 3$.

2.3.4 Real Estate House Price Prediction (KNN Regression)**Example 2.3.4: Real Estate Price Estimation (KNN Regression)**

A real estate agency uses KNN regression to estimate house prices based on:

- Area (in hundreds of sq. ft., e.g., 1400 sq. ft. $\implies$ 14)
- Age (in years)

Real Estate Housing Dataset

House	Area (hundred sq. ft.)	Age (years)	Price (Rs. Lakhs)
H_1	12	5	50
H_2	15	3	65
H_3	8	10	30
H_4	20	2	90
H_5	10	8	38
H_6	18	6	75

Task: Predict the price of a new house with Area = 14 (1400 sq. ft.) and Age = 4 years, i.e., $\mathbf{X} = (14, 4)$, using $k = 3$ and Manhattan distance.

Solution:

Step 1: Compute Manhattan distances from query house $\mathbf{X}(14, 4)$:

$$\begin{aligned} \text{To } H_1(12, 5) : d &= |14 - 12| + |4 - 5| = 2 + 1 = 3 \\ \text{To } H_2(15, 3) : d &= |14 - 15| + |4 - 3| = 1 + 1 = 2 \\ \text{To } H_3(8, 10) : d &= |14 - 8| + |4 - 10| = 6 + 6 = 12 \\ \text{To } H_4(20, 2) : d &= |14 - 20| + |4 - 2| = 6 + 2 = 8 \\ \text{To } H_5(10, 8) : d &= |14 - 10| + |4 - 8| = 4 + 4 = 8 \\ \text{To } H_6(18, 6) : d &= |14 - 18| + |4 - 6| = 4 + 2 = 6 \end{aligned}$$

Step 2: Sorted Manhattan distance ranking:

House	Distance	Price (Rs. Lakhs)
H_2	2	65
H_1	3	50
H_6	6	75
H_4	8	90
H_5	8	38
H_3	12	30

Step 3: Average prices across $k = 3$ nearest neighbors:

$$\begin{aligned} \{k = 3\} &= \{H_2, H_1, H_6\} \\ \hat{y} &= \frac{65 + 50 + 75}{3} = \frac{190}{3} \approx 63.33 \text{ Lakhs} \end{aligned}$$

$\therefore$ The new point $\mathbf{X} = (14, 4)$ is predicted to have a price of **Rs. 63.33 Lakhs** for $k = 3$.

2.4 Parameter Selection: Small k vs. Large k

Selecting the hyperparameter k balances local sensitivity against global regularization:

Table 2.2: Comparison of Small k versus Large k Characteristics

Aspect	Small k	Large k
Neighbourhood	Local	Broad
Sensitivity to Noise	High	Lower
Variance	Higher	Lower
Bias	Lower	Higher
Decision Boundary	Irregular / complex	Smoother
Risk	Overfitting	Underfitting

Guiding Principle for Choosing k

Choose k to balance bias and variance.

Chapter 3

Naive Bayes Classification

3.1 Probability Foundations & Notations

The Naive Bayes classifier is grounded in classical axiomatic probability theory and conditional distributions.

Table 3.1: Probability Notations and Terminology

Notation	Meaning
$P(A)$	Probability of event A
$P(A^c)$	Probability of complement of A
$P(A \cap B)$	Joint probability of A and B occurring simultaneously
$P(A \cup B)$	Probability of either event A or event B occurring
$P(A B)$	Conditional probability of event A given event B has occurred
$P(A, B)$	Joint probability of events A and B (alternative standard notation)

3.1.1 Key Probability Identities

- **Complement Rule:**

$$P(A^c) = 1 - P(A) \quad (3.1)$$

- **Addition Rule (Union):**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (3.2)$$

- **Joint Probability Equivalence:**

$$P(A \cap B) = P(A, B) \quad (3.3)$$

3.1.2 Conditional Probability & Multiplication Rule

Definition 3.1.1: Conditional Probability

For two events A and B with $P(B) > 0$:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (3.4)$$

Re-arranging yields the general **Multiplication Rule**:

$$P(A, B) = P(A | B)P(B) \quad (3.5)$$

$$P(A, B) = P(B | A)P(A) \quad (3.6)$$

3.1.3 Statistical Independence

If events A and B are statistically independent:

$$P(A \cap B) = P(A) \cdot P(B) \quad (3.7)$$

and equivalently:

$$P(A | B) = P(A) \quad (3.8)$$

3.2 Bayes' Theorem

Suppose an analyst wishes to determine the probability that an observation belongs to class C_k after observing feature evidence vector $\mathbf{X}$:

Theorem 3.2.1: Bayes' Theorem

$$P(C_k | \mathbf{X}) = \frac{P(\mathbf{X} | C_k)P(C_k)}{P(\mathbf{X})} \quad (3.9)$$

In conceptual terms:

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}} \quad (3.10)$$

3.3 Worked Numerical Example: Credit Risk Classification

Example 3.3.1: Credit Risk Classification

A retail lending institution assesses credit risk with two target states:

- C_1 : Good Credit
- C_2 : Default

Historical portfolio priors and likelihoods for a financial indicator vector $\mathbf{X}$ are given by:

$$\begin{aligned} P(C_1) &= 0.80, & P(C_2) &= 0.20 \\ P(\mathbf{X} | C_1) &= 0.10, & P(\mathbf{X} | C_2) &= 0.60 \end{aligned}$$

Task: Compute the evidence $P(\mathbf{X})$, determine the posterior probabilities $P(C_1 | \mathbf{X})$ and $P(C_2 | \mathbf{X})$, and classify the borrower.

Solution:

Step 1: Compute Total Evidence $P(\mathbf{X})$ via Law of Total Probability:

$$\begin{aligned} P(\mathbf{X}) &= P(\mathbf{X} | C_1)P(C_1) + P(\mathbf{X} | C_2)P(C_2) \\ &= (0.10 \times 0.80) + (0.60 \times 0.20) \\ &= 0.08 + 0.12 \\ &= 0.20 \end{aligned}$$

Step 2: Calculate Posterior Probabilities:

$$P(C_1 | \mathbf{X}) = \frac{P(\mathbf{X} | C_1)P(C_1)}{P(\mathbf{X})} = \frac{0.10 \times 0.80}{0.20} = \frac{0.08}{0.20} = 0.40$$

$$P(C_2 | \mathbf{X}) = \frac{P(\mathbf{X} | C_2)P(C_2)}{P(\mathbf{X})} = \frac{0.60 \times 0.20}{0.20} = \frac{0.12}{0.20} = 0.60$$

Step 3: Comparison and Decision:

$$P(C_2 | \mathbf{X}) > P(C_1 | \mathbf{X}) \quad (0.60 > 0.40)$$

$\therefore$ Classify $\mathbf{X}$ as **Default** (C_2).

3.4 The Bayes Classifier Framework

3.4.1 Marginal Evidence Across K Classes

For a classification problem with K discrete classes $C_1, C_2, \dots, C_K$, the overall marginal probability of evidence $\mathbf{X}$ is:

$$P(\mathbf{X}) = \sum_{k=1}^K P(\mathbf{X} | C_k)P(C_k) \quad (3.11)$$

3.4.2 Optimal Bayes Decision Rule

The Bayes classifier assigns an observation $\mathbf{X}$ to the class maximizing the posterior probability:

$$\hat{c} = \arg \max_{C_k} P(C_k | \mathbf{X}) = \arg \max_{C_k} \frac{P(\mathbf{X} | C_k)P(C_k)}{P(\mathbf{X})} \quad (3.12)$$

Because the evidence denominator $P(\mathbf{X})$ is identical across all classes C_k , the decision rule simplifies to:

$$\hat{c} = \arg \max_{C_k} P(\mathbf{X} | C_k)P(C_k) \quad (3.13)$$

3.5 The Naive Bayes Classifier

3.5.1 The Challenge of High-Dimensional Likelihoods

When an observation is characterized by p features $\mathbf{X} = (x_1, x_2, \dots, x_p)$:

$$P(C_k | x_1, x_2, \dots, x_p) = \frac{P(x_1, x_2, \dots, x_p | C_k)P(C_k)}{P(x_1, x_2, \dots, x_p)} \quad (3.14)$$

Directly estimating the joint multivariate conditional distribution $P(x_1, x_2, \dots, x_p | C_k)$ from finite sample sizes is computationally and statistically challenging due to exponential data sparsity.

3.5.2 The Naive Conditional Independence Assumption

Definition 3.5.1: The Naive Assumption

All input features $x_1, x_2, \dots, x_p$ are mutually conditionally independent given the class label C_k :

$$P(x_1, x_2, \dots, x_p \mid C_k) = \prod_{j=1}^p P(x_j \mid C_k) \quad (3.15)$$

that is:

$$P(x_1, x_2, \dots, x_p \mid C_k) = P(x_1 \mid C_k) \cdot P(x_2 \mid C_k) \cdots P(x_p \mid C_k) \quad (3.16)$$

3.5.3 Naive Bayes Classification Rule

Applying the independence assumption produces the core operational classification formula:

$$\hat{c} = \arg \max_{C_k} P(C_k) \prod_{j=1}^p P(x_j \mid C_k) \quad (3.17)$$

Why Is It Termed “Naive”?

The assumption is termed “naive” because conditional independence among predictors is rarely strictly true in real-world scenarios.

Financial Example: In credit scoring, variables such as *Income*, *Debt*, and *Employment Status* clearly exhibit mutual dependencies. However, in practice, the naive approximation dramatically simplifies computation while delivering strong empirical classification performance.

3.5.4 Algorithmic Step-by-Step Sequence

Naive Bayes Execution Procedure

- i) **Start with Training Data:** Assemble feature vectors $\mathbf{X}_i$ and class labels y_i .
- ii) **Calculate Prior Probabilities:** Compute class base rates $P(C_k) = \frac{N_{C_k}}{N}$.
- iii) **Calculate Univariate Likelihoods:** Estimate 1-D conditional probabilities $P(x_j | C_k)$ for each feature given each class.
- iv) **Apply Naive Assumption:** Decompose joint likelihood into the product:

$$P(x_1, \dots, x_p | C_k) = \prod_{j=1}^p P(x_j | C_k)$$

- v) **Calculate Class Score:** Evaluate the unnormalized joint score for each candidate class:

$$\text{Score}(C_k) = P(C_k) \prod_{j=1}^p P(x_j | C_k)$$

- vi) **Compare Scores Across All Classes:** Evaluate $\text{Score}(C_k)$ across $k = 1, \dots, K$.
- vii) **Predict Output Class:**

$$\hat{c} = \arg \max_{C_k} \text{Score}(C_k)$$

Chapter 4

Logistic Regression

4.1 Introduction and Core Motivation

Logistic regression is a fundamental supervised learning algorithm utilized for classification problems. Unlike regression models that predict a continuous numerical quantity, logistic regression models the conditional probability that a given observation belongs to a particular class.

4.1.1 Why Not Standard Linear Regression?

In ordinary linear regression, the relationship between features and the target is modeled as:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon \quad (4.1)$$

Under this linear formulation, the predicted value $\hat{Y}$ can take any real value across the entire unbounded interval:

$$\hat{Y} \in (-\infty, +\infty) \quad (4.2)$$

However, probabilities are mathematically constrained to lie strictly within the unit interval:

$$P(Y = 1 \mid X) \in [0, 1] \quad (4.3)$$

If ordinary linear regression were used for classification, predictions could yield nonsensical probabilities less than 0 or greater than 1. Consequently, a transformation is needed that “squashes” the unbounded linear output $(-\infty, +\infty)$ into the valid probability range $[0, 1]$. In logistic regression, this transformation is provided by the **sigmoid (logistic) function**.

4.2 The Sigmoid (Logistic) Function and Probability Modeling

Logistic regression first computes a linear combination of the input features:

$$Z = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \quad (4.4)$$

The real-valued score Z is then passed through the sigmoid (logistic) activation function to yield the predicted probability of the positive class ($Y = 1$):

$$P(Y = 1) = \frac{1}{1 + e^{-Z}} \quad (4.5)$$

4.2.1 Algebraic Equivalence of the Sigmoid Function

Multiplying both the numerator and the denominator by e^Z (since $\frac{e^Z}{e^Z} = 1$) provides the alternate standard representation:

$$P(Y = 1) = \frac{1 \cdot e^Z}{(1 + e^{-Z}) \cdot e^Z} \quad (4.6)$$

$$= \frac{e^Z}{e^Z + e^{-Z} \cdot e^Z} \quad (4.7)$$

$$= \frac{e^Z}{e^Z + e^{Z-Z}} \quad (4.8)$$

$$= \frac{e^Z}{e^Z + e^0} \quad (4.9)$$

$$= \frac{e^Z}{e^Z + 1} = \frac{e^Z}{1 + e^Z} \quad (4.10)$$

Substituting the linear combination $Z = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$ directly yields:

$$P(Y = 1) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}} \quad (4.11)$$

4.2.2 Properties of the Sigmoid Curve and Decision Threshold

The sigmoid curve possesses distinctive geometric characteristics:

- **S-Shaped Geometry:** The curve exhibits a smooth, monotonic S-shape.
- **Asymptotic Bounds:** As $Z \rightarrow -\infty$, $P(Y = 1) \rightarrow 0$. As $Z \rightarrow +\infty$, $P(Y = 1) \rightarrow 1$.
- **Symmetric Midpoint:** At $Z = 0$, $e^0 = 1$, yielding exactly $P(Y = 1) = \frac{1}{1+1} = 0.5$.

To convert continuous probability estimates into discrete class labels, a **decision threshold** is established (conventionally set to 0.5):

$$\hat{Y} = \begin{cases} 1, & \text{if } P(Y = 1) \geq 0.5 \quad (Z \geq 0) \\ 0, & \text{if } P(Y = 1) < 0.5 \quad (Z < 0) \end{cases} \quad (4.12)$$

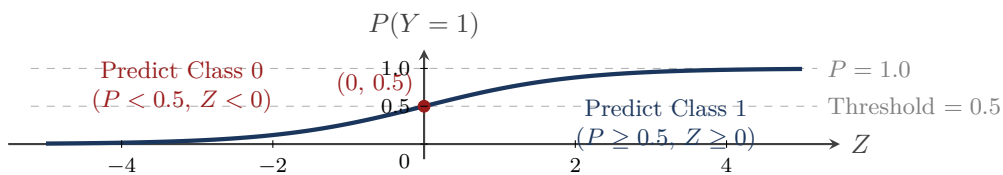


Figure 4.1: The Sigmoid activation function squashing unbounded linear values Z into valid class probabilities $[0, 1]$ with a decision boundary at $Z = 0$ ($P = 0.5$).

4.3 Key Terms and Matrix Notation

Definition 4.3.1: Core Variables of Logistic Regression

- **Independent Variables (X):** The set of input features or predictor variables $X_1, X_2, \dots, X_p$.
- **Dependent Variable (Y):** The categorical target outcome, typically binary $Y \in \{0, 1\}$.
- **Logistic Function (Sigmoid Function):** The nonlinear squashing function $\sigma(Z) = \frac{1}{1+e^{-Z}}$ mapping real scores to $(0, 1)$.
- **Coefficient (β_j):** The parameter estimated by the model for predictor X_j , indicating how strongly X_j affects the dependent variable on the log-odds scale.
- **Intercept (β_0):** The constant baseline term representing the log-odds when all independent variables equal zero ($X_1 = X_2 = \dots = X_p = 0$).

4.3.1 Matrix Representation

In compact linear algebra notation, the dataset of n observations and p features is represented as an $n \times p$ feature matrix X :

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} \quad (4.13)$$

where:

- n is the total number of observations (samples).
- p is the total number of features (predictors).

The observed class labels form a binary response vector $y \in \{0, 1\}^n$:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad y_i \in \{0, 1\} \quad (4.14)$$

4.4 Odds, Log-Odds (Logit), and Parameter Interpretation**4.4.1 Definition of Odds**

The **odds** of an event represent the ratio of the probability that the event occurs to the probability that the event does not occur:

$$\text{Odds} = \frac{P(Y = 1)}{1 - P(Y = 1)} \quad (4.15)$$

Example 4.4.1: Calculating Probabilities and Odds

Consider a basket containing 3 apples and 5 oranges (total items = $3 + 5 = 8$).

1. Probability of selecting an Orange:

$$P(\text{Orange}) = \frac{5}{8} = 0.625 \quad (4.16)$$

2. Odds of selecting an Orange:

$$\text{Odds(Orange)} = \frac{\text{Number of Oranges}}{\text{Number of Not Oranges (Apples)}} = \frac{5}{3} \approx 1.667 \quad (4.17)$$

3. Odds of selecting an Apple:

$$\text{Odds(Apple)} = \frac{\text{Number of Apples}}{\text{Number of Not Apples (Oranges)}} = \frac{3}{5} = 0.600 \quad (4.18)$$

4.4.2 Log-Odds (Logit Function)

Taking the natural logarithm of the odds produces the **log-odds** (or **logit**):

$$\ln(\text{Odds}) = \ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) \quad (4.19)$$

Substituting $P(Y=1) = \frac{e^Z}{1+e^Z}$ and $1-P(Y=1) = \frac{1}{1+e^Z}$:

$$\frac{P(Y=1)}{1-P(Y=1)} = \frac{\frac{e^Z}{1+e^Z}}{\frac{1}{1+e^Z}} = e^Z \quad (4.20)$$

Taking the natural logarithm:

$$\ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \ln(e^Z) = Z = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \quad (4.21)$$

Thus, **logistic regression is a linear model for the log-odds**.

4.4.3 Interpreting Coefficients and Odds Ratios

For a one-unit increase in predictor X_j (holding all other variables constant), the log-odds changes by β_j :

$$\Delta \ln(\text{Odds}) = \beta_j \quad (4.22)$$

Exponentiating both sides yields the **Odds Ratio (OR)**:

$$\text{Odds Ratio} = e^{\beta_j} \quad (4.23)$$

The directional impact of β_j on the odds is summarized as follows:

- $\beta_j > 0 \implies e^{\beta_j} > 1$: The odds of the event increase.
- $\beta_j < 0 \implies e^{\beta_j} < 1$: The odds of the event decrease.
- $\beta_j = 0 \implies e^{\beta_j} = 1$: Predictor X_j has no effect on the odds.

Example 4.4.2: Odds Ratio Interpretation with $\beta_j = 0.7$

Suppose the estimated coefficient for a predictor X_j is $\beta_j = 0.7$.

$$\text{Odds Ratio} = e^{\beta_j} = e^{0.7} \approx 2.0137 \approx 2 \quad (4.24)$$

Interpretation: A one-unit increase in X_j roughly doubles the odds of the outcome event occurring ($Y = 1$).

4.5 Model Training via Maximum Likelihood Estimation (MLE)

Logistic regression estimates its parameters $\beta_0, \beta_1, \dots, \beta_p$ using **Maximum Likelihood Estimation (MLE)**.

- **Objective of MLE:** Find the parameter vector β that maximizes the joint likelihood of observing the given sample data.
- **Lack of Closed-Form Solution:** Unlike ordinary least squares (OLS) regression where $\hat{\beta} = (X^T X)^{-1} X^T y$, there is **no closed-form analytical solution** for the MLE of logistic regression parameters.
- **Numerical Optimization:** Parameter estimates are solved numerically using iterative optimization algorithms (e.g., Newton-Raphson, Iteratively Reweighted Least Squares, or Gradient Ascent).

4.6 Assumptions of Logistic Regression

Valid inference and stable performance in logistic regression rely on six foundational assumptions:

1. **Independent Observations:** Sample observations must be independent of one another (no repeated measures or auto-correlation).
2. **Binary Dependent Variable:** The response variable Y is binary (for standard binomial logistic regression).
3. **Linearity in Log-Odds:** Independent variables exhibit a strictly linear relationship with the log-odds $\ln\left(\frac{P}{1-P}\right)$.
4. **Absence of Extreme Outliers:** The predictor space should not contain extreme leverage points or anomalous outliers.
5. **Adequately Large Sample Size:** Because MLE relies on large-sample asymptotic properties, sufficient sample size is required to prevent biased parameter estimates.
6. **Low Multicollinearity:** Predictor variables should exhibit little to no multicollinearity. High correlation among predictors inflates parameter standard errors.

4.7 Variants, Metrics, and Trade-Offs

4.7.1 Types of Logistic Regression

Depending on the nature of the target variable, logistic regression is categorized into three types:

- **Binomial Logistic Regression:** The target has exactly two classes (e.g., Yes / No, Default / Non-Default).
- **Multinomial Logistic Regression:** The target has three or more unordered nominal classes (e.g., Cat, Dog, Sheep).
- **Ordinal Logistic Regression:** The target has three or more ordered categories (e.g., Low, Medium, High).

4.7.2 Model Evaluation Metrics

Model performance for logistic regression is assessed using standard classification metrics:

- **Accuracy:** Ratio of correctly classified instances to total instances.
- **Precision:** Ratio of true positives to all predicted positives ($\frac{TP}{TP+FP}$).
- **Recall (Sensitivity):** Ratio of true positives to all actual positives ($\frac{TP}{TP+FN}$).
- **F1-Score:** Harmonic mean of precision and recall ($2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$).
- **AUC-ROC:** Area Under the Receiver Operating Characteristic curve, measuring discrimination across all possible classification thresholds.

4.7.3 Advantages and Disadvantages

Advantages (Pros)	Disadvantages (Cons)
• Outputs well-calibrated probabilities, not just hard class labels. • Highly interpretable parameters via odds ratios (e^{β_j}). • Directly extends to multiclass classification problems.	• Interaction terms between features must be constructed manually. • Interpretations are multiplicative on odds, not linear on probabilities. • Complete separation prevents convergence (coefficients diverge to $\pm\infty$). • Requires regularization (e.g., L1/L2 penalties) or informative priors to handle separation.

4.8 End-to-End Workflow and Generalized Linear Model (GLM) Classification

4.8.1 Five-Step Implementation Pipeline

1. **Prepare Data:** Encode categorical predictors (e.g., one-hot or dummy encoding) and scale continuous features if regularization is utilized.
2. **Fit Model:** Estimate parameter vector β using Maximum Likelihood Estimation (MLE) via numerical optimization.
3. **Interpret Coefficients:** Exponentiate parameters to interpret predictor effects as odds ratios (e^{β_j}).
4. **Predict and Threshold:** Compute predicted probabilities $P(Y = 1) = \sigma(X\beta)$ and assign class labels based on a selected threshold (typically 0.5).
5. **Evaluate Model:** Measure performance using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.

4.8.2 Membership in Generalized Linear Models (GLM)

Logistic regression is a member of the broader **Generalized Linear Models (GLM)** family. In GLM theory, a model is defined by three components:

1. **Random Component:** The dependent variable Y follows a Bernoulli (or Binomial) distribution: $Y \sim \text{Bernoulli}(p)$.

2. **Systematic Component:** A linear combination of predictors $Z = \beta_0 + \sum_{j=1}^p \beta_j X_j$.
3. **Link Function:** The **logit link function** $g(p) = \ln\left(\frac{p}{1-p}\right) = Z$, which maps the expected probability parameter from $[0, 1]$ to the unbounded real line $(-\infty, +\infty)$.